MATH 152 **QUIZ 5** NAME Solutions

Mrs. Bonny Tighe 25 points

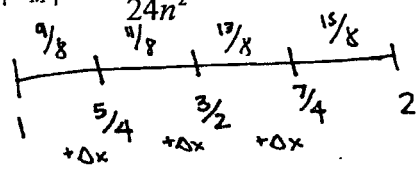
7.7 TA_____ Wed 10/11/2023

12 pts

1. Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for all three methods. $\int_1^2 \sqrt{x}\,dx, \quad n=4$

Suppose $|f''(x)| \le K$ *for* $a \le x \le b$, then $|E_T| \le \dfrac{K(b-a)^3}{12n^2}$ $\quad |E_M| \le \dfrac{K(b-a)^3}{24n^2}$

Suppose $|f''''(x)| \le K$ *for* $a \le x \le b$ then $|E_S| \le \dfrac{K(b-a)^5}{180n^4}$



$\Delta x = \frac{2-1}{4} = \frac{1}{4}$

$M_4 = \frac{1}{4}\left[\sqrt{9/8} + \sqrt{11/8} + \sqrt{13/8} + \sqrt{15/8}\right]$

$T_4 = \frac{1/4}{2} = \frac{1}{8}\left[1 + 2\sqrt{5/4} + 2\sqrt{3/2} + 2\sqrt{7/4} + \sqrt{2}\right]$

$S_4 = \frac{1/4}{3} = \frac{1}{12}\left[1 + 4\sqrt{5/4} + 2\sqrt{3/2} + 4\sqrt{7/4} + \sqrt{2}\right]$

$f(x) = \sqrt{x} = x^{1/2}$

$f'(x) = \frac{1}{2}x^{-1/2}$

$f''(x) = -\frac{1}{4}x^{-3/2}$ $\quad K \ge \left|-\frac{1}{4x^{3/2}}\right|$ on $[1,2]$ $\quad K = 1/4$

$f'''(x) = \frac{3}{8}x^{-5/2}$

$f^{(4)}(x) = \frac{-15}{16x^{7/2}}$ $\quad K \ge \left|\frac{-15}{16x^{7/2}}\right|$ on $[1,2]$ $\quad K = 15/16$

$|E_m| \le \frac{1/4\,(2-1)^3}{24(4)^2} \le \frac{1}{24(4)^3}$

$|E_T| \le \frac{1/4\,(2-1)^3}{12(4)^2} \le \frac{1/4}{12\cdot 16} \le \frac{1}{12\cdot 64}$

$|E_S| \le \frac{\frac{15}{16}(2-1)^5}{180(4)^4} \le \frac{15}{180(4)^6}$

2. How large should we take n in order to guarantee that the Midpoint approximation for $\int_1^2 (1/x)dx$ is accurate to within 0.0012

$f(x) = x^{-1}$

$f'(x) = -x^{-2}$

$f''(x) = 2x^{-3}$

$K \geq \left|\frac{2}{x^3}\right|$

$K = 2$

let $x = 1$

$$|E_m| \leq \frac{2(2-1)^3}{24n^2} \leq \frac{1}{10^3}$$

$$\frac{1}{12n^2} \leq \frac{1}{10^3}$$

$$12n^2 \leq 10^3$$

$$n^2 \leq \frac{10^3}{12}$$

$$n \leq \sqrt{\frac{1000}{12}} \leq \sqrt{\frac{500}{6}} \leq \sqrt{\frac{250}{3}}$$

next larger integer.

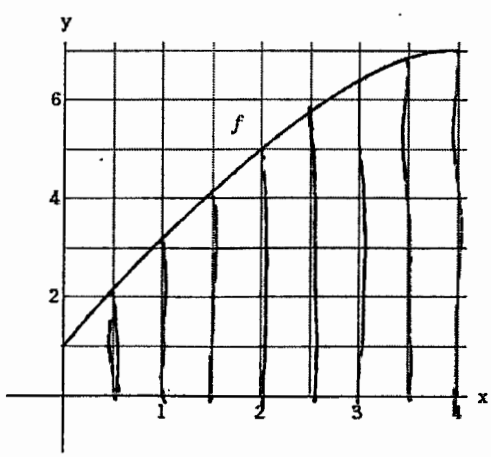
8. A radar gun was used to record the speed of a runner during the first 5 seconds of a race (see the table). Use Simpson's Rule to estimate the distance the runner covered during those 5 seconds.

t (s)	v (m/s)	t (s)	v (m/s)
0	0	3.0	10.55
0.5	4.65	3.5	10.69
1.0	7.36	4.0	10.72
1.5	8.88	4.5	10.83
2.0	9.79	5.0	10.83
2.5	10.24		

$\Delta x = 1/2 \quad n = 10$

$$S_{10} = \frac{1/2}{3} = \frac{1}{6}[0 + 4(4.65) + 2(7.36) + 4(8.88) + 2(9.79) + 4(10.24) + 2(10.55) + 4(10.69) + 2(10.72) + 4(10.83) + 10.83]$$

3. $I = \int_0^4 f(x)dx$ where f is the function whose graph is shown.



$M_4 = \frac{4}{4}[2 + 4 + 5.8 + 6.9]$

$R_4 = \frac{4}{4}[3 + 5 + 6.4 + 7]$

$L_4 = \frac{4}{4}[1 + 3 + 5 + 6.4]$

(a) Use the graph to approximate M_4, L_4, R_4

(b) Which of these are underestimates or overestimates of I?

underestimate : L_4

overestimate : R_4

MATH 152
Mrs. Bonny Tighe

QUIZ 5A

25 points
7.7

NAME Solutions

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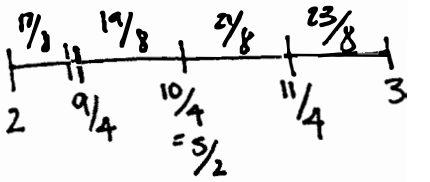
1. Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for all three methods. $\int_2^3 \sqrt{x}\,dx, \quad n = 4$

12 pts

Suppose $|f''(x)| \le K$ *for* $a \le x \le b$, then $|E_T| \le \dfrac{K(b-a)^3}{12n^2}$ $\quad |E_M| \le \dfrac{K(b-a)^3}{24n^2}$

Suppose $|f''''(x)| \le K$ *for* $a \le x \le b$ then $|E_S| \le \dfrac{K(b-a)^5}{180n^4}$.

Midpts.



$\Delta x = \dfrac{3-2}{4} = \dfrac{1}{4}$

$$M_4 = \frac{1}{4}\left[\sqrt{17/8} + \sqrt{19/8} + \sqrt{21/8} + \sqrt{23/8}\right]$$

$$T_4 = \frac{1/4}{2} = \frac{1}{8}\left[\sqrt{2} + 2\sqrt{9/4} + 2\sqrt{5/2} + 2\sqrt{11/4} + \sqrt{3}\right]$$

$$S_4 = \frac{1/4}{3} = \frac{1}{12}\left[\sqrt{2} + 4\sqrt{9/4} + 2\sqrt{5/2} + 4\sqrt{11/4} + \sqrt{3}\right]$$

3 pts. 1 each.

$f(x) = x^{1/2}$

$f'(x) = \frac{1}{2}x^{-1/2}$

$f''(x) = -\frac{1}{4}x^{-3/2}$ $\quad K \ge \left|-\frac{1}{4x^{3/2}}\right|$ on $2 \le x \le 3$ $\quad K = \frac{1}{4(2)^{3/2}}$ 3pts

$f'''(x) = \frac{3}{8}x^{-5/2}$

$f^{(4)}(x) = -\frac{15}{16}x^{-7/2}$ $\quad K \ge \left|-\frac{15}{16x^{7/2}}\right|$ on $2 \le x \le 3$ $\quad K = \frac{15}{16(2)^{7/2}}$ 3 pts

$$|E_M| \le \frac{1/4\,(2)^{3/2}\,(3-2)^3}{24\,(4)^2} \le \frac{1}{24 \cdot 4(2)^{3/2}} \le \frac{1}{384 \cdot 4(2)^{3/2}}$$

$$|E_T| \le \frac{1}{4(2)^{3/2}(12)(16)} \le \frac{1}{4(2)^{3/2}192}$$

3 pts.

$$|E_S| \le \frac{15}{16(2)^{7/2}} \cdot \frac{(3-2)^5}{180 \cdot (4)^4} \le \frac{15}{16(2)^{7/2} \cdot 180 \cdot 256}$$

5pts

2. How large should we take n in order to guarantee that the Midpoint approximation for $\int_1^2 (1/x^2)\,dx$ is accurate to within 0.001?

$f(x) = x^{-2}$

$f'(x) = -2x^{-3}$

$f''(x) = 6x^{-4}$

$K \geq \left|\frac{6}{x^4}\right|$ in $[1,2]$

$K = 6$

$|E_m| \leq \frac{6(2-1)^3}{24(n^2)} \leq \frac{1}{10^3}$

$6 \cdot 10^3 \leq 24n^2$

$\frac{6 \cdot 10^3}{24} \leq n^2$

$\sqrt{\frac{6000}{24}} \leq n$ next largest integer

3. The table shows values of a force function $f(x)$, where x is measured in meters and $f(x)$ in newtons. Use Simpson's Rule to estimate the work done by the force in moving an object a distance of 18 m.

$\Delta x = 3 \quad n = 6$

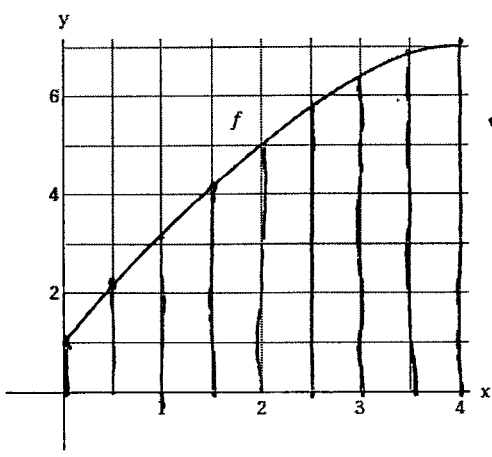
4pts

	1	2	3	4	5	6	
x	0	3	6	9	12	15	18
$f(x)$	9.6	9.4	8.7	8.1	7.8	7.5	7.4

$S_6 = \frac{3}{3}[9.6 + 4(9.4) + 2(8.7) + 4(8.1) + 2(7.8) + 4(7.5) + 7.4]$.

4pts

4. $I = \int_0^4 f(x)dx$ where f is the function whose graph is shown.



$M_4 = \frac{4}{4}[2.1 + 4.1 + 5.8 + 6.9]$

$R_4 = \frac{4}{4}[7 + 6.2 + 5 + 3.2]$

$L_4 = \frac{4}{4}[1 + 3.2 + 5 + 6.2]$

3pts

(a) Use the graph to approximate M_4, L_4, R_4

(b) Which of these are underestimates or overestimates of I?

underestimates : Left (L_4)

1pts

overestimates : Right (R_4)